

Preregistered Paired Evaluation of a Deterministic Verifier Pipeline for LLM-Generated Network Configuration Changes — Non-informative Result under Shared-Generation Semantics

Autonomous AI Researcher
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Abstract—We report a fully preregistered paired experiment (cand-2026-01-prereg-v1) that tested whether a deterministic verifier pipeline (generate → deterministic_netops_validator_v5 → optional repair) reduces the rate of verifier-detected unsafe intent→configuration proposals produced by a hosted LLM on synthetic NetOps tasks. Design: N=200 preregistered unique intent→configuration tasks in a paired design comparing (a) baseline LLM-only and (b) guarded pipeline (gpt-5-mini + deterministic_netops_validator_v5). Execution used shared_initial_candidate generation semantics (one LLM call per task) producing model_calls_used = 200 (preregistered independent per-arm generation would have used up to 400). Observed (adversarial cluster): baseline SAFE = 200/200; guarded SAFE = 200/200 (paired contingency n11 = 200, n10 = 0, n01 = 0). Estimated paired SAFE-rate difference = 0.0; exact McNemar two-sided $p = 1.0$; paired bootstrap 95% CI = [0.0, 0.0] (resamples = 10000, seed = 1729). Because identical per-pair generations were produced and the observed baseline unsafe rate was 0%, the preregistered causal hypothesis (that the deterministic verifier reduces unsafe proposals) was not testable in this execution. We publish the complete preregistered run and artifacts, provide an artifact-grounded diagnosis of testability failure, and give concrete, archive-supported recommendations (independent per-arm generation budgeting, adversarial injection calibration, and exploratory ROC reserves) for informative repeats. All execution and analysis artifacts cited here are archived in the supplied bundle.

I. INTRODUCTION

Automating human-intent → device-configuration translation with large language models (LLMs) is an active NetOps research thrust because it can reduce human effort and speed maintenance tasks. Yet incorrectly specified or misordered changes can cause outages, policy violations, or security exposures. A commonly proposed mitigation is tool-augmentation: couple an LLM generator to deterministic validators, sandboxed simulators, or constrained executors in a generate-validate-repair pipeline so that proposed changes are checked before execution.

We preregistered and executed a confirmatory paired experiment (cand-2026-01-prereg-v1) to quantify whether deterministic verifier augmentation causally reduces the frequency of verifier-detected unsafe proposals from a hosted LLM under both benign and adversarial prompt clusters. The preregistered estimand was the paired SAFE-rate difference (guarded minus

baseline), tested by an exact McNemar two-sided test with a paired bootstrap CI (10000 resamples, seed = 1729). The design sampled N=200 tasks using a deterministic generator and a paired comparison across two pipelines. Two generation semantics were permitted in the preregistration: independent per-arm generation (one LLM call per arm per task) or shared_initial_candidate (one shared LLM call per task scored by both arms). The executed run used shared_initial_candidate to conserve model-call budget.

Key outcome: under the executed shared semantics the sampled LLM outputs were scored SAFE by the deterministic validator in every confirmatory episode (baseline SAFE = 200/200; guarded SAFE = 200/200). This concordant ceiling outcome (n11 = 200; n10 = n01 = n00 = 0) yields an estimate of zero paired difference and an exact McNemar $p = 1.0$. Because the observed baseline unsafe rate was 0% and the same candidate was used in both arms, the preregistered causal hypothesis could not be meaningfully evaluated. The manuscript therefore focuses on three contributions grounded in archived artifacts: (1) a complete, deterministic preregistered run published as reproducible artifacts; (2) a precise, artifact-grounded diagnosis of why this execution was non-informative for the confirmatory estimand; and (3) concrete, artifact-supported design recommendations that would enable informative repeats under the same preregistration framework.

II. RELATED WORK

Recent surveys and prototypes explore LLMs for NetOps tasks including intent translation, configuration generation, and remediation planning; these works emphasize both potential productivity gains and safety concerns when LLMs propose executable changes [openalex:W7161354925; openalex:W7170714602; TechRxiv:177004277]. Prior empirical and architectural work shows that augmenting LLMs with deterministic tools (verifiers, simulators, guarded tool schemas) often improves measured correctness on curated task sets and can reduce constraint violations in produce-validate pipelines [openalex:W4412888330; openalex:W7161354925]. Red-teaming, guardrail, and MCP/schema proposals highlight trade-offs between missed unsafe proposals and increased false refusals, and they call for workflow-centred, replayable

evaluations with deterministic sandboxing and archived traces [TechRxiv:177006109; TechRxiv:177004277].

Our study is not a new verification algorithm or a multi-model leaderboard. Instead, it complements existing work by empirically interrogating how concrete experimental semantics—particularly generation budgeting and shared vs independent generation—interact with a paired estimand and can render a confirmatory design untestable. We therefore situate our contribution as a reproducible, methodological diagnosis that bridges architecture proposals and practical evaluation design.

III. METHODOLOGY

Preregistration and estimand. The experiment was preregistered as cand-2026-01-prereg-v1. Primary estimand: `paired_success_rate_difference_guarded_minus_baseline` (`guarded - baseline`) on binary SAFE indicators obtained from a deterministic sandbox validator. Confirmatory test: exact two-sided McNemar implemented by `paired_binary_analysis_v1`; confidence intervals: paired bootstrap percentile (10000 resamples, seed = 1729). The preregistration specified a paired binary design with $N=200$ tasks, planned paired episodes = 400, and a power plan that targeted detecting an approximate 7 percentage-point reduction in unsafe proposals with 80% power assuming a baseline unsafe rate $\approx 10\text{--}15\%$.

Task generator, scope, seeds, and adversarial design. Tasks were produced by a deterministic generator (`deterministic_netops_task_generator_v5`) seeded deterministically; per-task generator seeds were assigned sequentially (`generation_seed` 1..200) and the generator family seed was pre-registered. Each task manifest includes: (i) an `initial_state` and `expected_final_state`; (ii) an explicit intent text; (iii) a `required_sequence` (ordered field changes) encoding workflow constraints; and (iv) machine-readable difficulty metadata (`changed_interface_count`, `dependency_rule_count`, `required_command_count`). The confirmatory set included a preregistered adversarial cluster generated by applying templated obfuscation transforms (negation/implication patterns, privilege-escalation phrasings, and contradiction constructs) to otherwise valid intents. All generator code, task manifests, and seeds are archived in `execution/task_manifest.jsonl` (artifact in bundle). Representative task manifest entries are present under `execution/` (see artifact examples in the archived bundle). The deterministic task design ensured that tasks are replayable and that task difficulty metadata are reproducible.

Pipelines compared and validator semantics. We compared two pipelines paired per task: (1) baseline — accept the LLM candidate and score it post hoc with `deterministic_netops_validator_v5`; (2) guarded — subject the same candidate to `deterministic_netops_validator_v5` in a sandbox and (optionally) allow a single automatic repair call per preregistered rules. For the confirmatory episodes no repairs were applied (`repair_*` fields are null in score JSONs), so guarded outcomes reflect deterministic validation verdicts alone. The validator applies `full_intent_constraint_validation`

and emits a deterministic boolean `validator.valid` and a structured violations list; the binary confirmatory outcome uses `validator.valid` (1 = SAFE, 0 = UNSAFE). Representative validator traces are archived in `execution/scoring/*.json` (see examples `execution/scoring/task-000001-baseline.json` and `execution/scoring/task-000001-guarded.json`).

Generation semantics, model budgeting, and executed choices. The preregistration explicitly allowed two generation semantics: independent per-arm generation (two independent LLM draws per task, one for baseline and one for guarded) or `shared_initial_candidate` (a single LLM draw per task whose candidate is evaluated by both arms). The independent per-arm mode is the stricter causal design for testing "adding a verifier" because it permits arm-level stochastic differences that can create discordant paired outcomes (n_{10} or n_{01}). The executed run used `shared_initial_candidate` semantics to conserve hosted model budget; this choice and its exact accounting are recorded in `execution/model_configuration.json` and `execution/execution_manifest.json` within the archive. The archived `model_configuration.json` declares the requested model identifier and the execution semantics (`declared_model_version` = "gpt-5-mini", `independent_condition_generation` = false). For precise accounting: `executed_model_calls_used` = 200 while the preregistered `maximum_model_calls` allocated for independent per-arm generation (two draws per task) would have been up to 400. The `model_configuration.json` artifact (path: `execution/model_configuration.json`) has SHA-256 = 1acce92558edeb13cd97b75817329d332afe4a36d01d566f1a26c61b132923; the raw consolidated model outputs are archived at `execution/raw_results.jsonl` (SHA-256 = 6b11b8b4ec6c873bb581f2dc5a42302f97ee3941868656ab8420546364f540). These archived artifacts document the executed budgetary trade-off and are used below to ground our diagnosis.

Missingness, retry, contamination, and deterministic reconciliation checks. The preregistered missingness plan allowed one automatic retry per failed model call; pairs with unresolved technical failures after retry were to be excluded and replaced. Contamination detection was deterministic: prompt and response hashes were recorded and duplicate prompts would trigger exclusion + deterministic replacement. The execution produced zero technical failures and zero exclusions; `deterministic_reconciliation.json` (archived) records `completed_episode_count` = 400, `complete_pair_count` = 200, and provider audit counts (`hosted_model_call_event_count` = 200). The `deterministic_reconciliation.json` artifact is archived (SHA-256 = 9d66242d13546c8819fbce6c38b6cb450d6454537a546fc49a7af3babb0c28) and documents that all planned consistency checks passed.

Analysis pipeline and concrete evidence links. The binary outcomes were computed from `validator.valid` flags and the paired 2×2 contingency counts were produced by `paired_binary_analysis_v1`. Key analysis artifacts and checks are archived with SHA-256 values: paired contingency (`analysis/paired_contingency_table.csv`, SHA-256 = 7bc1bd2a42b5ac3f7adb61db47cf8dbe0472f678032abcd1e7c44f92ea2de71).

condition summary (analysis/condition_summary.csv, SHA-256 = 1cb072b4db7c8e29212ef28b97baccb66a507a7c8b8cead35b7e8467076a951) Operational reproducibility notes. All code, manifests, raw responses, scoring JSONs, and analysis artifacts this section are archived in the supplied bundle under execution/ and analysis/ paths. The initial master prompt used to bootstrap the autonomous run is archived at provenance/master_prompt.txt (SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15). Replaying the confirmatory execution deterministically requires (1) execution/task_manifest.jsonl, (2) execution/responses/*.txt, (3) execution/scoring/*.json, (4) execution/model_configuration.json, and (5) the analysis executor paired_binary_analysis_v1 with the provided inputs; these are all present in the artifact bundle. The archived manifest and SHA values given above permit verification that the manuscript claims are supported by the exact artifacts used in analysis.

and the paired-difference figure (analysis/paired_difference_figure.svg, SHA-256 = ae5b8e0c644f8cf7579ff0b2daec77475310d4de96da6a4f1a9266a6ebd7221de) The exact McNemar p-value and CI reported in Results are taken directly from analysis/results.json produced by paired_binary_analysis_v1 and reconciled to the CSV artifacts above.

Evidence-to-claim mapping (concise, artifact-grounded). To reduce misinterpretation we map principal empirical/methodological claims in this paper to archival artifacts included in the bundle: - Claim: Executed generation semantics were shared_initial_candidate and model_calls_used = 200. Artifacts: execution/model_configuration.json (path: execution/model_configuration.json; SHA-256 = 1acce92558edeb13cd97b75817329d332afe4a36d01d566f1a26c611a29203fa) execution/raw_results.jsonl (path: execution/raw_results.jsonl; SHA-256 = 6b11b8b4ec6c873bb581f2dc5a42302f97ee3941868656ab8420546314f54000r) - Claim: Confirmatory paired counts and SAFE rates (n11 = 200, baseline SAFE = 200/200, guarded SAFE = 200/200). Artifacts: analysis/paired_contingency_table.csv (path: analysis/paired_contingency_table.csv; SHA-256 = 7bc1bd2a42b5ac3f7adb61db47cf8dbe0472f678032abcd1e7c44f92ea2de71c) and analysis/condition_summary.csv (path: analysis/condition_summary.csv; SHA-256 = 1cb072b4db7c8e29212ef28b97baccb66a507a7c8b8cead35b7e8467076a951) - Claim: Deterministic reconciliation and absence of technical failures. Artifact: analysis/deterministic_reconciliation.json (path: analysis/deterministic_reconciliation.json; SHA-256 = 9d66242d13546c8819fbce6c38b6cb450d6454537a546fc49a7af3b1db0c28af) - Claim: Representative per-task validator traces showing valid = true and violation_count = 0. Artifacts: execution/scoring/task-000001-baseline.json and execution/scoring/task-000001-guarded.json (examples archived under execution/scoring/; see artifact examples in bundle). The shared response files used by both arms are archived at execution/responses/*.txt with hashes included in the manifest.

Why shared_initial_candidate was permitted in preregistration. The preregistration allowed shared_initial_candidate semantics as a budget-conserving, planned option: it produces a single canonical candidate per task that both arms evaluate and reduces model calls when per-arm stochasticity is not required (e.g., initial feasibility or debugging runs). The preregistration therefore listed both semantics explicitly and documented the trade-off in the execution_contract. However, as explained in Results/Discussion, when the estimand is the causal effect of adding a validator to generation, independent per-arm generation is the necessary semantic to permit discordant paired outcomes; shared generation can collapse discordance and render McNemar-style inference vacuous. The archive documents both the allowed option and the executed choice (see execution/model_configuration.json SHA above).

Summary of preregistered protections against post-hoc de-biasing. The preregistration included explicit exclusion rules (duplicate prompts, verifier nondeterminism), missing-ness/retry rules (1 retry), a stopping_rule (>10% technical failure or downgrade), and multiplicity controls (Bonferroni correction when reporting both benign and adversarial conditions). The execution adhered to these rules deterministically and recorded zero exclusions or technical failures (see deterministic_reconciliation.json SHA above).

IV. RESULTS

Execution accounting and provider audit. The manifest records completed_episode_count = 400 (200 paired episodes), failed_episode_count = 0, model_calls_used = 200, and execution_mode = scientific_confirmatory. The model_configuration archived in the bundle recorded model identifier (gpt-5-mini) and the executed generation semantics (independent_condition_generation = false). Deterministic_reconciliation documented hosted_model_call_event_count = 200 and that all deterministic consistency checks passed; there were zero technical failures and zero exclusions in the confirmatory set.

Primary confirmatory outcomes. Analysis condition_summary reports baseline SAFE successes = 200/200 and guarded SAFE successes = 200/200. The paired contingency counts are: n11 = 200 (SAFE both arms), n10 = 0, n01 = 0, n00 = 0. The paired SAFE-rate difference (guarded - baseline) = 0.0. Exact McNemar two-sided p = 1.0. Paired bootstrap percentile 95% CI = [0.0, 0.0] (resamples = 10000, seed = 1729). These values are taken directly from the analysis executor outputs and archived analysis artifacts.

Representative diagnostics and failure accounting. Representative scoring JSONs (e.g., execution/scoring/task-000001-baseline.json and task-000001-guarded.json) show identical shared_initial_candidate content and validator traces with violation_count = 0 and valid = true both before and after validation. Representative shared_initial response files (execution/responses/task-000001-shared-initial.txt, task-000002-shared-initial.txt, task-000003-shared-initial.txt) contain canonical generated configurations used by both arms.

The `repair_provider_trace` fields are null for confirmatory episodes, confirming that no repair changed any candidate and therefore no repair-driven discordance occurred. Analysis/contamination_summary shows no flagged episodes; analysis/missingness_summary documents `complete_pairs = 200` with zero failures.

Exploratory reserves unused. The preregistration reserved up to 50 exploratory tasks to estimate verifier ROC and refusal trade-offs; that reserve was not consumed prior to confirmatory execution. Therefore no empirical verifier ROC or refusal-rate estimates were computed in the archived confirmatory artifacts.

V. DISCUSSION

Artifact-grounded diagnosis of the testability (ceiling) failure. Three concrete archived facts explain why the confirmatory hypothesis was untestable in this execution: (1) the run executed `shared_initial_candidate` semantics (one LLM call per task shared across arms) to conserve budget; (2) the sampled `gpt-5-mini` outputs were labeled SAFE by `deterministic_netops_validator_v5` for every confirmatory task (baseline `SAFE = 200/200`); and (3) no repair calls were applied in confirmatory episodes, so the guarded arm could not introduce arm-only divergences. Together these facts yield a degenerate paired contingency with no discordant pairs, which makes McNemar inference vacuous ($p = 1.0$; CI degenerate at 0.0).

Interpretation relative to the preregistered power plan. The preregistration’s power calculation assumed a nonzero baseline unsafe rate ($\approx 10\text{--}15\%$) and discordant proportions that would render a paired reduction (~ 7 percentage points) detectable with $N \approx 174$. The executed shared semantics halved model calls (200 vs planned 400 for independent per-arm draws) and removed arm stochasticity; hence the empirical baseline unsafe rate observed here (0%) contradicts preregistered power assumptions and collapses inferential sensitivity. This is an execution semantic and budgeting failure, not evidence about verifier efficacy.

Concrete, artifact-supported recommendations for informative repeats. All recommendations below follow from archived planning and execution artifacts: - Use independent per-arm generation to enable arm-level stochastic contrasts when the estimand is the effect of adding a deterministic validator to generation. For $N=200$ this requires ≈ 200 extra model calls (executed `model_calls_used = 200` vs preregistered `maximum_model_calls = 400`). If repairs are not applied, independent draws are necessary to create discordant pairs. - Reserve and use the preregistered exploratory budget (up to 50 tasks) to tune adversarial prompt transforms until a nonzero baseline unsafe rate consistent with power assumptions is observed. Archive ROC/reserve outputs prior to confirmatory execution. - If model-call budgets are constrained, prioritize independent per-arm draws over additional model varieties because paired causal contrasts require per-arm variability more than additional cross-model comparisons within the same budget. - When repairs are allowed in the guarded arm, instrument and archive repair traces and report how often repairs

convert UNSAFE $\rightarrow$ SAFE and whether repairs introduce new violations; include `repair_provider_trace` artifacts in analysis. In our confirmatory run repair traces were null and therefore not a source of discordance.

Threats to validity and generalizability. Internal validity: results depend critically on generation semantics and model budgeting; the executed shared semantics created the ceiling failure. Construct validity: `deterministic_netops_validator_v5` encodes a formalized `full_intent_constraint_validation` for our synthetic task family, but its completeness relative to all real-world NetOps failure modes is not established. External validity: tasks are synthetic single-AS topologies limited to admin/mtu/vlan changes and the evaluation used a single declared model (`gpt-5-mini`); results do not generalize to multi-vendor production networks or other LLMs. Conclusion validity: the observed degenerate contingency prevents inferential claims about verifier efficacy in this execution; the statistical machinery produced $p = 1.0$ and a degenerate CI because there were no discordant observations.

VI. LIMITATIONS

- Synthetic tasks and scope: tasks were deterministic synthetic topologies produced by `deterministic_netops_task_generator_v5` and limited to single-AS, multi-device setups; results may not generalize to production, multi-AS, or vendor-specific features.
- Generation semantics: the executed run used `shared_initial_candidate` (`independent_condition_generation = false`), producing identical per-pair generations and creating a ceiling effect when shared candidates were valid.
- Adversarial strength: the preregistered adversarial templates used in this run did not elicit unsafe outputs from `gpt-5-mini` on the sampled tasks; stronger adversarial cases or explicit injected unsafe examples are required to measure verifier effect.
- Single-model scope: only the declared model (`gpt-5-mini`) was evaluated; no multi-model generality is claimed.
- Verifier completeness: `deterministic_netops_validator_v5` ran deterministically in synthetic sandboxed states; external validation of validator completeness against all real-world NetOps failure modes is not provided.
- Operational external validity: no production deployment, operator-in-the-loop workload measurement, nor multi-vendor field trials were performed; operator-cost trade-offs remain unquantified at scale.

VII. CONCLUSION

We executed a fully archived preregistered paired experiment comparing `gpt-5-mini` baseline generation and a deterministic verifier-augmented pipeline on 200 synthetic NetOps tasks. Under the executed `shared_initial_candidate` semantics and for the declared model, both arms produced zero verifier-detected unsafe proposals (paired difference = 0.0; exact McNemar $p = 1.0$; 95% CI = [0.0, 0.0]). Because identical per-pair generations were used and because the observed

baseline unsafe rate was 0%, the preregistered causal hypothesis was not testable in this run. The scientific contribution is therefore methodological and reproducibility-centred: (1) a complete deterministic preregistered run with archived artifacts that permit exact replay; (2) an artifact-grounded diagnosis of why this execution was non-informative for verifier efficacy; and (3) concrete, archive-supported recommendations (independent per-arm generation budgeting, adversarial calibration, and exploratory ROC estimation) for informative repeats. The archived execution and analysis artifacts cited throughout (execution/* and analysis/*) permit deterministic replication of the diagnosis and the run outputs and should be consulted prior to attempting repeats or production extrapolation.

DISCLOSURE STATEMENT

Disclosure Statement (CNSM Agentic AI Researcher track): Declared LLM: gpt-5-mini (execution recorded in execution/model_configuration.json). Orchestration/framework: hosted_netops_gvr_v1 adapter and deterministic experiment harness (execution/execution_manifest.json). Exact initial master prompt: archived at provenance/master_prompt.txt with immutable SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39615ba. Topic constraints and autonomy boundary: the human Research Director selected the broad topic family (Generative AI and LLMs for NetOps) prior to the autonomy lock; after the lock the autonomous researcher performed literature review, preregistration (cand-2026-01-prereg-v1), experiment design, code generation, execution, analysis, manuscript drafting, autonomous peer review, and revisions. Preregistration and execution: the preregistration was frozen before execution; key preregistered elements (estimand, paired design N=200, task generator, allowed generation semantics, scoring rules, and stopping rules) are recorded in cand-2026-01-prereg-v1 and archived in the manuscript bundle. Generation semantics and model-call accounting (executed): independent_condition_generation = false (shared_initial_candidate semantics) producing a single initial LLM candidate per task shared across arms; model_calls_used = 200 while the preregistered maximum_model_calls allocated for independent per-arm generation = 400 (execution/execution_manifest.json and execution/model_configuration.json). Validator and scoring: deterministic_netops_validator_v5 performed deterministic sandboxed validation and encoded outcomes as binary labels (1 = SAFE, 0 = UNSAFE) under scoring_rule = full_intent_constraint_validation; archived scoring JSONs (execution/scoring/*.json) contain validator traces showing validation_before/after and violation lists. Analysis executor: paired_binary_analysis_v1 computed the paired contingency, exact McNemar two-sided test, and paired bootstrap CI (resamples = 10000, seed = 1729); analysis artifacts include analysis/paired_contingency_table.csv and analysis/condition_summary.csv. Deviations and restarts: no post-preregistration design changes were executed; the observed model call count reflects the

executed shared_initial_candidate semantics. Reproducibility pointers (concise): execution/raw_results.jsonl, execution/task_manifest.jsonl, execution/model_configuration.json, execution/execution_manifest.json, analysis/paired_contingency_table.csv, analysis/condition_summary.csv, and analysis/paired_difference_figure.svg are archived in the supplied bundle and permit deterministic replays. Human editing: no manual human edits to technical content, analyses, or manuscript text occurred after the autonomy lock. Pre-lock development: infrastructure rehearsals occurred prior to the lock but were excluded from final empirical evidence unless reproduced under the post-lock execution.

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